

ADE or CoNLL04 sentence

Published tokens to source character offsets
No annotation-guided rescue windows

Dataset-specific training KG

Typed mentions, directed edges, provenance
Train-derived labels and endpoint constraints

Provenance-isolated context construction

Exact matching + frozen BGE-M3 relation-pattern retrieval
Clear graph hints for exact train/evaluation text overlaps

SOE: QLoRA adapter

Source only; no KG
Frozen Qwen3-4B
Structured candidate JSON

EAE: QLoRA adapter

Exact concept hints
Frozen Qwen3-4B
Structured candidate JSON

HRGE: QLoRA adapter

Anchors / edges / patterns
Frozen Qwen3-4B
Structured candidate JSON

Span reconstruction

Copied text to exact offsets
Resolve relation endpoints

Span reconstruction

Copied text to exact offsets
Resolve relation endpoints

Span reconstruction

Copied text to exact offsets
Resolve relation endpoints

SOE comparison

Not used in PGE fusion

Entity acceptance: ANY

EAE agreement OR anchor
OR verified endpoint
Deduplicate accepted entities

Relation gate: ALL

References; type signatures;
explicit status; evidence
Output: EVGE relations

CFE validation ablation

Accepted entities + raw HRGE relations
Retain only surviving endpoints

PGE evidence-linked assertion graph

Accepted entities + EVGE relations
Retain only surviving endpoints

Audit records accompany predictions

Source spans, relation direction, provenance and acceptance/rejection signals
Rules establish structural admissibility, not semantic truth or clinical causality